

# Empirical Evidence Concerning AES

PETER HELLEKALEK and STEFAN WEGENKITTL

University of Salzburg

---

AES, the Advanced Encryption Standard, is one of the most important algorithms in modern cryptography. Certain randomness properties of AES are of vital importance for its security. At the same time, these properties make AES an interesting candidate for a fast nonlinear random number generator for stochastic simulation. In this article, we address both of these two aspects of AES. We study the performance of AES in a series of statistical tests that are related to cryptographic notions like confusion and diffusion. At the same time, these tests provide empirical evidence for the suitability of AES in stochastic simulation. A substantial part of this article is devoted to the strategy behind our tests and to their relation to other important test statistics like Maurer's Universal Test.

Categories and Subject Descriptors: E.3 [Data Encryption]: Standards; G.3 [Probability and Statistics]: Random number generation; I.6.0 [Simulation and Modeling]: General

General Terms: Algorithms, Experimentation, Security

Additional Key Words and Phrases: AES, block ciphers, diffusion, confusion, statistical tests

---

## 1. INTRODUCTION

The Advanced Encryption Standard (AES) is a new cryptographic algorithm that will play a central role in securing electronic communication in the next decades. It became a Federal Information Processing Standard on November 26, 2001, see FIPS PUB 197 [NIST 2001a].

AES was selected as the official successor to DES. It is a block cipher, hence a special case of a symmetric cipher system. For these cryptographic notions, we refer the reader to the comprehensive Handbook of Applied Cryptography [Menezes et al. 1997]. AES encrypts (decrypts) successive large blocks of plaintext (ciphertext) in the same key-dependent manner. Both the sender and the receiver of encrypted messages have to use the same key  $k$ , which has to be kept secret. The security of such a "secret key" algorithm depends on the

---

Research for this work has been supported by the Austrian Science Fund (FWF), project no. S8303-MAT and by the Austrian National Bank, Jubiläumsfondsprojekt 7576.

Authors' address: Institut für Mathematik, Universität Salzburg, Hellbrunnerstr. 34, A-5020 Salzburg, Austria; email: Peter.Hellekalek@sbg.ac.at.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or direct commercial advantage and that copies show this notice on the first page or initial screen of a display along with the full citation. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers, to redistribute to lists, or to use any component of this work in other works requires prior specific permission and/or a fee. Permissions may be requested from Publications Dept., ACM, Inc., 1515 Broadway, New York, NY 10036 USA, fax: +1 (212) 869-0481, or [permissions@acm.org](mailto:permissions@acm.org).

© 2003 ACM 1049-3301/03/1000-0322 \$5.00

amount of computational power that is necessary to decrypt the ciphertext without knowing the key.

AES was designed to fulfill several criteria, in particular, (a) speed and compactness on a wide range of platforms (AES is a fast algorithm, efficient enough to be used in stochastic simulations), and (b) resistance against all known attacks (AES breaks up patterns that might exist in the plaintext).

Statistical testing of AES is interesting in several respects. Concerning security, one should not be able to detect any systematic correlations between ciphertext and plaintext blocks by statistical analysis. With respect to stochastic simulation, one would like to know how AES performs as a random number generator. AES is a very interesting candidate for a fast nonlinear pseudorandom number generator for stochastic simulation which is not likely to suffer from simple intrinsic structures like linear generators. With properties such as high speed, nonlinearity, thorough theoretical analysis, and portability, AES appears to be an attractive addition not only to applied cryptography but also to the simulationist's toolbox of advanced RNGs. There is, currently, no nonlinear equivalent to the most prominent members L'Ecuyer et al. [2002a] and Matsumoto and Nishimura [1998] of this toolbox, which are both linear algorithms.

In this article:

- the standard output streams of AES are tested, where both parameters, plaintexts and keys are varied,
- state-of-the-art statistical tests, that have proved to be among the most stringent test designs to assess uniform random numbers are employed,
- our selection of tests and test parameters is justified,
- the performance of Maurer's Universal Test for Random Bit Generators [Maurer 1992] in comparison to advanced versions of the classical serial test [Knuth 1998] is discussed.

It appears that our test results are the first detailed statistical results for AES in the published literature.

## 2. BACKGROUND INFORMATION

We refer the reader to the monograph of Menezes et al. [1997] for general concepts of cryptography, like the notion of a *block cipher*, and to Daemen and Rijmen [2002] for details on AES.

AES is the result of an international competition to select the successor to DES. The winning algorithm was *Rijndael*, which had been proposed by Daemen and Rijmen. The only difference between AES and Rijndael is the range of supported values for the block and the key lengths (see Daemen and Rijmen [2002, Chap. 3]).

AES is an *iterated* block cipher. The ciphertext block is the result of a number of key-dependent Boolean transformations applied to the plaintext block.

Boolean transformations are Boolean functions that map binary vectors to binary vectors of the same length.

Like all modern ciphers, AES has been constructed with two basic design principles in mind, *confusion* and *diffusion*.

Confusion is a term used by Shannon [1949] to denote the lack of information concerning the dependence of the key on the plaintext and the ciphertext. No statistical analysis of the ciphertext should enable an attacker to gain substantial information on the key, even if the cipher algorithm is known by the attacker.

Diffusion is also meant to frustrate a statistical analysis (see Shannon [1949]). It denotes the quantitative spreading of information. For any key, no statistical dependence between simple structures in the plaintext and simple structures in the ciphertext should exist.

Like any other block cipher, AES may be used in different *modes of operation*. In the simplest one, the Electronic Code Book mode (ECB), each plaintext block is encrypted independently, using the same key. Given a sequence  $p_1, p_2, \dots$  of plaintext blocks, a key  $k$ , and an encryption function  $e_k$ , then the sequence of ciphertext blocks  $c_1, c_2, \dots$  in ECB mode is the sequence  $e_k(p_1), e_k(p_2), \dots$ .

In the Output Feedback mode (OFB), the block cipher is used as a *synchronous stream cipher*. We choose some initial value (i.e., input block)  $z_0$  and then generate a sequence of pseudorandom blocks of bits by repeated application of the encryption function  $e_k$ :

$$z_i = e_k^{(i)}(z_0), \quad e_k^{(i)} := \underbrace{e_k \circ e_k \circ \dots \circ e_k}_{i \text{ times}}.$$

The sequence  $(z_i)_{i \geq 0}$  is then used as the key stream for the Vernam cipher  $c_i = p_i \oplus z_i, i \geq 0$ . Here,  $\oplus$  denotes addition modulo 2. The sequence  $(z_i)_{i \geq 0}$  acts as a one-time pad.

In the Counter mode, we first generate a key stream  $(z_i)_{i \geq 0}$  in the following manner. We encrypt an incrementing counter (or some other predictable sequence) by  $e_k$ . Hence, if  $x_0, x_1, \dots$  is the sequence of blocks that are generated by our counter, then the key stream  $(z_i)_{i \geq 0}$  is defined as  $z_i = e_k(x_i)$ . In the second step, the plaintext is encrypted by the Vernam cipher  $c_i = p_i \oplus z_i, i \geq 0$ .

In the PRNG mode, in the initialization step, the cipher key  $k$  is extracted from some *entropy pool* (see Menezes et al. [1997] for this concept). After this step, a sequence of pseudorandom bit blocks  $(z_i)_{i \geq 0}$  is generated by repeated application of AES to some initial value  $z_0$ , like in the OFB mode.

If we want to generate random numbers for stochastic simulations, the counter mode seems preferable to the PRNG mode. It guarantees full period length (in our case, a period length of  $2^{128}$  for the output sequence), whereas there are no guarantees available for the period length in PRNG mode. Furthermore, the counter mode allows the direct generation of a particular element of its output sequence, without the need to compute any predecessors. We would like to point out that for the Rijndael algorithm, even longer period lengths up to  $2^{256}$  are available (see Daemen and Rijmen [2002, Chap. 3.1]).

AES was tested as follows. The block length for plaintext and key was set to 128 bits. The different modes of AES were studied with four different setups:

- (1) RNG stands for a setup where AES is acting as a random number generator, like in the OFB or PRNG modes. The initial state  $z_0$  as well as the key  $k$  were set to all-zero. The output stream  $(z_i)_{i \geq 0}$  was the sequence  $z_i = e_k^{(i)}(z_0)$ .
- (2) DIFF denotes a test designed to assess diffusion. The key  $k$  is set to all-zero and a highly patterned plaintext is encrypted. The sequence  $(c_i)_{i \geq 0}$  of ciphertext blocks is then tested for randomness. This kind of setup goes back to Spencer and Travares [1984]. Each 128-bit plaintext block contained exactly 6 ones, which were distributed systematically among the 128 bits. The first plaintext was 1111110...0, the second was 11111010...0, the 123rd was 111110...01, followed by 11110110...0, and so on.
- (3) PCOUNT denotes a counter mode. The sequence  $(p_i)_{i \geq 0}$  of plaintext blocks was generated by computing the binary representation of the integer  $i$  and padding with zeroes. The key was set to all-zero. The output stream was  $(c_i)_{i \geq 0}$ , with  $c_i = e_k(p_i)$ .
- (4) KCOUNT is a setup that uses the same counter as PCOUNT, with the roles of plaintext and key interchanged. Hence, the output stream  $(c_i)_{i \geq 0}$  was generated by the rule  $c_i = e_{k_i}(p_0)$ .

Each of these test setups produces a stream of 128-bit blocks which is transformed into a stream of bits  $(y_i)_{i \geq 0}$ ,  $y_i \in \{0, 1\}$ , by concatenating the 128-bit words. The resulting bit stream is used as input in the statistical tests discussed in the next section.

### 3. STATISTICAL TESTS

Empirical (“black-box”) testing of a random number generator complements a priori theoretical analysis of its structural properties. The goal is to check for empirical evidence against the presumed hypothesis  $H_0$  that the random numbers are sampled from a uniformly distributed independent bit process. If no such evidence is found with reasonable effort, the numbers are said to pass the test. Statistical tests may thus be interpreted as sample applications. These applications may or may not suffer from the deterministic origin and the resulting regularities of the random numbers. In fact, statistical tests are often chosen such that their power is concentrated on some kind of alternative hypotheses that have been defined with the construction method of the pseudo-random numbers in mind. The term “black-box” testing is not adequate in this situation.

When testing statistical properties of block ciphers like AES, we face a somewhat different situation. The design of such algorithms excludes comprehensive theoretical analysis of the structures and regularities of their output. The situation closely resembles the assessment of hardware generators. We are checking for the presence of *any unwanted* structures, that is, patterns and correlations which can be described in easy mathematical terms. Thus, a statistical assessment of AES has to cover a broad range of alternative hypotheses where each test needs to be highly specific (no evidence against  $H_0$  is missed) while still

retaining a lot of sensitivity (no false rejections of  $H_0$ ). Several strategies have been proposed:

*Strategy I.* Use a broad range of test statistics which are specific to various defects. Examples are the DIEHARD battery of Marsaglia [1996], the forthcoming test package of L'Ecuyer and Simard [2002], and the NIST test battery [NIST 2001b]. AES has been evaluated using the NIST package (see NIST [2001a]). In principle, we consider this type of testing necessary and valuable. However, up to now, AES has only been tested using relatively small sample sizes. Furthermore, the exact test procedure and the parameters that were used have not been published. There are no refereed publications on this topic. In addition, it has to be pointed out that some of these test statistics give highly correlated results in the presence of low-dimensional defects (see Section 3.1). The reader should note that the correlations among the tests (the “inter-test” correlations) in the NIST battery have been analyzed with respect to a *good* source of randomness. It remains an open problem to carry out the same type of inter-test correlation analysis for a range of more or less defective pseudo-random number generators.

*Strategy II.* Use a statistical test which claims to find almost every possible defect in the output stream, like the Universal Test of Maurer [1992]. The generality of the test will require larger sample sizes than more specific tests in order to yield comparable  $p$ -values. In Section 3.1, we report on the practical value of the Universal Test. In fact, we have achieved better overall performance using the following test strategy.

*Strategy III.* Employ systematic testing using a series of standard serial tests on  $d$ -dimensional overlapping bit-vectors sampled from the generator. Vary  $d$  and the sample size as widely as possible and report the  $p$ -value for each combination of these parameters. Strategy III enables us to check explicitly for *any* systematic deviations from equidistribution in dimensions  $d \in \{1, 2, \dots, d_{\max}\}$ . In practice,  $d_{\max}$  is limited by the available computational resources. This kind of testing will be appropriate for applications which are not sensitive to correlations in dimensions higher than those that have been tested. Strategy III has been applied by L'Ecuyer et al. [2002b] and Leeb and Wegenkittl [1997] in connection with testing pseudorandom number generators used in stochastic simulation. We believe that it should also be part of every statistical assessment of cryptographic algorithms.

In Section 3.1, we report a sample study on the test statistics known as *approximate entropy*, *overlapping serial test* and *universal test*. We then specify our test procedure for AES and our strategy to extend the applicable range of dimensions  $d$  by a technique called *dimension reduction* in Section 3.2. Section 4 summarizes our results for AES.

In the following, we always consider a sample  $X = (x_i)_{i=0}^{m-1}$  of bits  $x_0, x_1, \dots, x_{m-1}, x_i \in \mathcal{A} = \{0, 1\}$ . Here,  $m > 1$  is the number of bits used in a test. Let  $n > 1$  be the sample size and denote by  $\tilde{x}_i^d$  and  $\bar{x}_i^d$ ,  $0 \leq i < n - 1$ , the overlapping and nonoverlapping  $d$ -tuples  $\tilde{x}_i^d = (x_i, x_{i+1}, \dots, x_{i+d-1})$  and  $\bar{x}_i^d = (x_{i-d}, x_{i-d+1}, \dots, x_{i-d+d-1})$ . In the latter definition, we let  $x_{i+n} := x_i$  for

$i = 0, 1, \dots, d - 1$ . Note that  $m = d \cdot n$  bits are required for the nonoverlapping sample, whereas only  $m = n$  bits are required in the overlapping case. Finally, if  $\mathbf{a} = (a_1, \dots, a_d) \in \{0, 1\}^d$ , we let  $\tilde{\pi}_{\mathbf{a}}^d = \frac{1}{n} \#\{0 \leq i < n : \tilde{x}_i = \mathbf{a}\}$  and  $\bar{\pi}_{\mathbf{a}}^d = \frac{1}{n} \#\{0 \leq i < n : \bar{x}_i = \mathbf{a}\}$  denote the relative frequencies of the  $d$ -tuples in the sample. All logarithms below are in base 2.

### 3.1 Inter-Test Correlations

A test for the randomness of the bits  $x_0, x_1, \dots, x_{m-1}$  is a test for the null-hypothesis  $H_0$  that the  $x_i$  are sampled from a uniform distribution on  $\mathcal{A}$ , independently of each other. In Maurer [1992], this hypothesis is contrasted by an alternative hypothesis  $H_1$ , that the  $x_i$  are sampled from a stationary ergodic source which is not i.i.d. uniform and can be modeled as a Markov chain of order  $\kappa$ . Among all these models, the maximal value 1 of the entropy  $H$  is achieved uniquely under  $H_0$ , so that entropy is a discriminating property.

An estimator (see Wegenkittl [2001a]) of the entropy of a stationary ergodic source is

$$\hat{H}_f^d = - \sum_{\mathbf{a} \in \mathcal{A}^d} \tilde{\pi}_{\mathbf{a}}^d \log \tilde{\pi}_{\mathbf{a}}^d + \sum_{\mathbf{a} \in \mathcal{A}^{d-1}} \tilde{\pi}_{\mathbf{a}}^{d-1} \log \tilde{\pi}_{\mathbf{a}}^{d-1},$$

which is based on the frequencies of  $d$ -dimensional overlapping tuples in the sample  $X$ . Following Maurer [1992], another estimator can be constructed from the return times of nonoverlapping tuples  $\tilde{x}_i^d$ ,  $\hat{H}_r^d = \frac{1}{d \cdot n} \sum_{i=Q}^{Q+n-1} \log T(i)$ . Here,  $T(i) = \min\{1 \leq j \leq i + 1 : \tilde{x}_{i-j}^d = \tilde{x}_i^d\}$  denotes the truncated return time to  $\tilde{x}_i^d$  which is always well defined and finite if we let  $\tilde{x}_{-1}^d := \tilde{x}_i^d$ . This definition allows a very efficient implementation of this test. The quantity  $Q$  denotes a “warm-up”. The test needs  $Q + n$  nonoverlapping tuples in total. In our examples, we always set  $Q = 10 \cdot 2^d$ , which is the value recommended in Maurer [1992]. As  $n$  tends to infinity,  $\hat{H}_f^d$  is an asymptotically unbiased consistent estimator for the entropy  $H$  of a stationary ergodic source, provided that the order  $\kappa$  of the source is smaller than  $d$  (see also Wegenkittl [2001a]). With this property, a statistical test based on  $\hat{H}_f^d$  can be used to check systematically for unwanted correlations up to dimension  $d$ . Thus, it qualifies for Strategy III. The estimator  $\hat{H}_r^d$  is an asymptotically unbiased consistent estimator for the expected logarithm of the return time to the initial  $d$ -tuple divided by  $d$ . As  $d$  approaches infinity, it also converges to the entropy  $H$ .

In order to compute  $p$ -values, the point-estimators  $\hat{H}_f^d$  and  $\hat{H}_r^d$  are usually scaled, yielding the following statistics and corresponding asymptotic distributions for large values of  $n$ :

$$\hat{I}^d = 2n(1 - \hat{H}_f^d) \xrightarrow{D} \chi_{2^d - 2^{d-1}}^2,$$

and

$$\hat{N}^d = \frac{\hat{H}_r^d - E[\cdot]}{\sqrt{V[\cdot]}} \xrightarrow{D} N[0, 1],$$

where  $\chi_l^2$  denotes a chi-square distribution with  $l$  degrees of freedom,  $N[0, 1]$  is a standard normal, and  $E[\cdot]$  and  $V[\cdot]$  denote expectation and variance of

the estimator  $\hat{H}_r^d$  (see Maurer [1992] and Coron and Naccache [1998] for tabulated values). In the literature,  $\hat{I}^d$  is also referred to as *approximate entropy* [Pincus and Singer 1998a, 1998b] and *generalized I-divergence* [Wegenkittl 2002]. The statistic  $\hat{N}^d$  is Maurer's Universal Test. A measure closely related to  $\hat{I}^d$  is the well-known overlapping serial test,

$$\hat{\chi}^d = n \sum_{\mathbf{a} \in \mathcal{A}^d} \frac{(\tilde{\pi}_{\mathbf{a}}^d - (1/2)^d)^2}{(1/2)^d} - n \sum_{\mathbf{a} \in \mathcal{A}^{d-1}} \frac{(\tilde{\pi}_{\mathbf{a}}^{d-1} - (1/2)^{d-1})^2}{(1/2)^{d-1}}.$$

Like  $\hat{I}^d$ ,  $\hat{\chi}^d$  is a Strategy III-test. It is able to find any deviation from equidistribution that can be modeled by an order  $\kappa$  ergodic stationary source provided that  $\kappa < d$ . In Wegenkittl [1999], the relationship between these measures is analyzed in the context of generalized  $\phi$ -divergences.

**3.1.1 Defective Source Analysis.** In Wegenkittl [2001a], the relationship of  $\hat{I}^d$  and  $\hat{N}^d$  with respect to ordinary (order one) Markov chains has been analyzed. Here we complement this by including an empirical study showing the ability of these statistics to detect an order  $\kappa \geq 1$  defect in a sample process  $X$  which is constructed in the following way. Let  $x_0, \dots, x_{\kappa-1}$  be random initial bits and, iteratively, let  $x_i = x_{i-\kappa}$  with probability  $\lambda$  and  $x_i = 1 - x_{i-\kappa}$  with probability  $1 - \lambda$ . In our study, the simulation of such a sequence uses the TT800 random number generator [Matsumoto and Kurita 1994]. Note that this process is i.i.d. uniform on  $\mathcal{A}$  if and only if  $\lambda = 1/2$ . In any other case, the entropy of the process is non-maximal, although the marginal distributions up to dimension  $\kappa$  are uniform. This model of a defective random source is interesting with respect to analyzing the adequateness of statistical tests for the assessment of block ciphers. Correlations introduced by  $\lambda \neq 1/2$  are “hidden” in higher dimensions.

As an example, let  $\lambda = 0.49$  so that the per-bit-entropy  $H$  of the process  $(x_i)_{i \geq 0}$  is  $H \approx 0.999711$ . Figure 1 shows gray scale plots of 16 repetitions of the 2-sided  $p$ -values of each statistic with a sample size  $n = 2^{16}$  and parameters  $\kappa \in \{1, \dots, 6\}$  and  $d \in \{1, \dots, 16\}$ . The  $p$ -values  $< 0.01$  (below the 1% level against  $H_0$ ) are denoted by black dots (small rectangles) in the plot. All other  $p$ -values have been plotted ranging from gray (the  $p$ -value is 1) to white (the  $p$ -value is 0.01).

Both statistics  $\hat{I}^d$  and  $\hat{\chi}^d$  can be seen to detect the defect at the 1% level of significance (black rectangles in plot) provided  $d > \kappa$  and provided that  $d$  is not too large. In fact, the results in dimensions  $d \leq 7$  differ only marginally. Consequently, the corresponding gray-values cannot be distinguished in the plots. This fact has already been observed in L'Ecuyer and Hellekalek [1998]. If  $d > 7$ , the sample size  $n$  is too small in comparison to the defect and the test makes systematic type-2 errors (not rejecting  $H_0$  although  $H_1$  applies). Note that both statistics use exactly  $n = 2^{16}$  bits to compute a single  $p$ -value. For  $d > 13$ ,  $\hat{I}^d$  correctly rejects  $H_0$ , but the reason is that due to the slow convergence to the asymptotic distribution, the test would actually reject any given sequence. It is not applicable any more. For the statistic  $\hat{\chi}^d$ , convergence is considerably better. Nevertheless, the defect is not detected in high dimensions

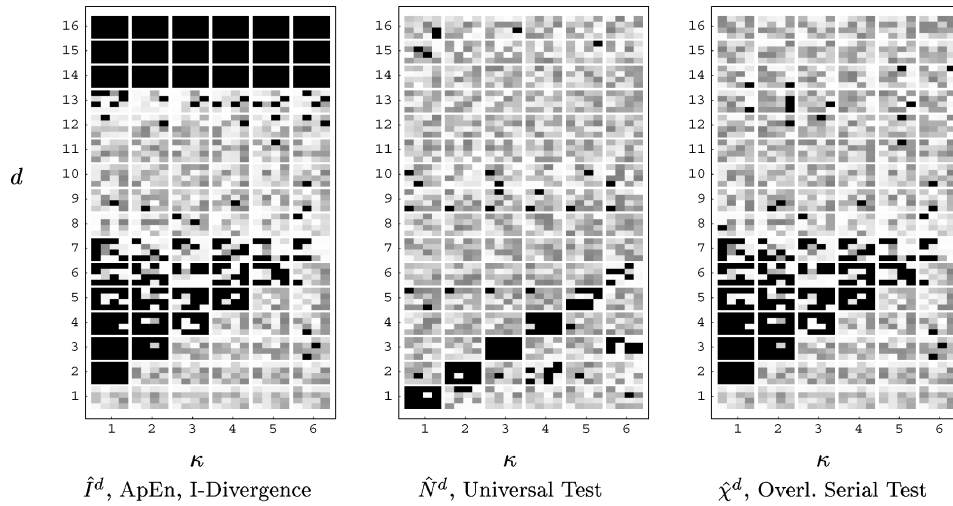
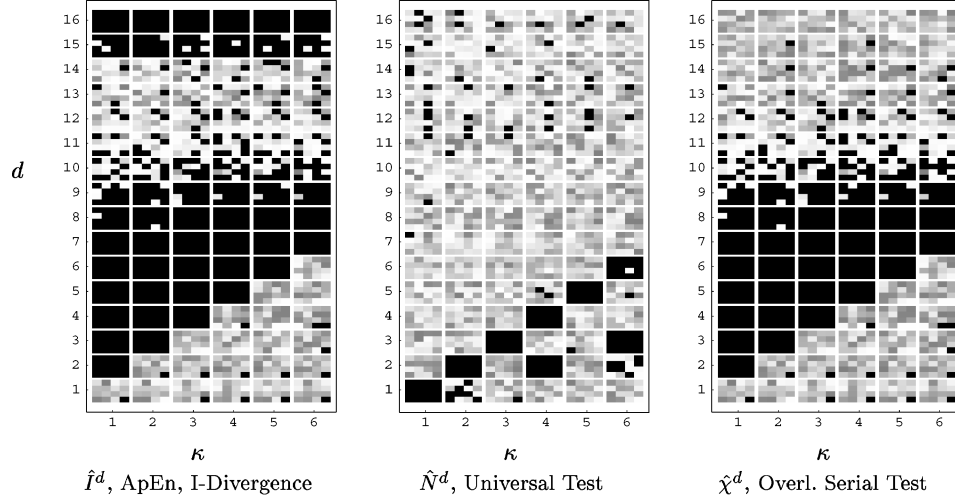


Fig. 1. Two-sided  $p$ -values of 16 independent repetitions (arranged in  $4 \times 4$  blocks) for  $d$  ranging from 1 (bottom line) to 16 (top line) and  $\kappa$  ranging from 1 (left column) to 6 (right column). Sample size  $n = 2^{16}$ , parameter  $\lambda = 0.49$ , entropy  $H \approx 0.999711$ . Black dots denote  $p$ -values smaller than 0.01.

$d > 13$  because the discriminating power of the test is wasted on too many alternatives. In summary, both statistics show highly correlated results. The defect is best detected if the design parameter  $d$  is a little larger than the order of the defect, which will be difficult to guess if one has no information on  $\kappa$ . Besides increasing the quality of the approximation by the asymptotic chi-square distribution, increasing the sample size  $n$  will also give less type-2 errors and correct rejection of  $H_0$  in dimensions  $d > 7$ , as can be seen in Figure 2, where  $n = 2^{18}$ .

In Figure 1, the statistic  $\hat{N}^d$ —despite claiming universality—is correctly rejecting  $H_0$  only if  $d = \kappa$ . If one increases the sample size  $n$ , the statistic is sometimes able to correctly reject  $H_0$  even if  $d < \kappa$  (see Figure 2). However, for a fair comparison, one has to be aware of the fact that  $\hat{N}^d$  needs up to 176 times the number of bits required by  $\hat{\chi}^d$  due to the use of nonoverlapping tuples and warm-up! The universal test based on return times needs much larger sample sizes to find low-dimensional defects. This is the price that has to be paid for being able to detect some high-dimensional deviations from uniformity with a test whose dimension is smaller than the order of the model for the source of randomness,  $d < \kappa$ .

We conclude that among the three candidates for statistical tests, the results are highly correlated in the presence of the kind of defects simulated in our study. We will consequently restrict ourselves to a single test and stress the variation of the parameter  $d$  and the sample size  $n$ . Clearly, the overlapping serial test is the most favorable candidate among the tests considered above since its asymptotics are well understood and easy to handle, and the discriminating power seems to be much better than that of the universal test.

Fig. 2. Same setup as in Figure 1 but sample size increased to  $n = 2^{18}$ .

### 3.2 Testing Procedure and Dimension Reduction

We apply two setups for testing the AES block cipher.

*Setup 1.* Vary  $d \in \{1, 2, 4, 8, 16\}$  and  $n \in \{2^{20}, 2^{21}, \dots, 2^{28}\}$ . Cut out every 8th bit of the stream  $(y_i)_{i \geq 0}$  starting with  $y_0$ , that is, set  $x_0 = y_0$ ,  $x_1 = y_8$ , and so on. For each combination of  $d$  and  $n$ , first compute 16 independent repetitions  $\hat{\chi}_1^d(d, n), \dots, \hat{\chi}_{16}^d(d, n)$  of  $\hat{\chi}^d$  using successive blocks of samples from an AES based generator. Then calculate one-sided  $p$ -values for the Kolmogorov-Smirnov test comparing the empirical distribution of the  $\hat{\chi}_i^d(d, n)$ ,  $1 \leq i \leq 16$ , to that of a chi-square distribution with  $2^d - 2^{d-1}$  degrees of freedom. This results in  $5 \times 9$   $p$ -values, one for each combination of  $d$  and  $n$ .

To check for the presence of high-dimensional correlations in AES, one would like to increase  $d$  to values such as 128 or 256. However, a brute-force implementation of such a test will lead to problems with storing the array of  $2^d$  counters for the calculation of the relative frequencies  $\tilde{\pi}_a^d$ . Two solutions have been proposed in literature. The first, so-called sparse tests (which can be implemented efficiently by Hash tables) use much more cells than samples ( $2^d \gg n$ ). Typically, one counts the number of cells that are hit at least once (or twice, or never, respectively). These tests have been shown to be highly powerful for detecting the regular structure of certain types of pseudorandom number generators used in stochastic simulation (see L'Ecuyer et al. [2002b]). It is not clear if this discriminating power also holds for the kind of defects that might be present in cryptographic generators.

In this article, we follow an idea introduced in Wegenkittl [2001b] and Wegenkittl and Matsumoto [1999] called dimension reduction (see also L'Ecuyer and Simard [1999] for a related test statistic). We map  $d$ -dimensional overlapping tuples to 3 distinct states as follows. Choosing  $d$  even and  $t = d/2$ , we count the number of 1's in the first  $d - 1$  coordinates of  $\tilde{x}_i^d$ . If this number is smaller than  $t$ ,  $\tilde{x}_i^d$  is mapped to state 1. Otherwise, the last coordinate  $x_{i+d-1}$

in  $\tilde{x}_i^d$  determines the state, which is set to  $x_{i+d-1} + 1$ . The three states here are motivated by a gambling game in Wegenkittl [2001b], the resulting test is, therefore, called *gambling test*. For each sample, the number of tuples  $\tilde{x}_i^d$  that is mapped to each of the three states is counted. A modified chi-square statistic,  $\hat{\chi}_g^d$ , say, is applied to the resulting three-dimensional counter vector. Again, the procedure is repeated 16 times and the empirical distribution of the resulting statistics is compared to a chi-square distribution with two degrees of freedom by means of a Kolmogorov–Smirnov test. We refer to Wegenkittl [2001b] and Wegenkittl and Matsumoto [1999] for details on the computation of the statistics and their distribution under  $H_0$ . In summary, we have the following:

*Setup 2.* Vary  $d \in \{32, 64, 128, 256\}$  and  $n \in \{2^{22}, \dots, 2^{28}\}$ . Compute 16 independent repetitions of the gambling test on the bit stream  $(y_i)_{i \geq 0}$  with parameters  $d$  and  $t = d/2$  and the 1-sided Kolmogorov–Smirnov  $p$ -value of their empirical distribution.

**3.2.1 Remarks.** The first setup stresses good equidistribution of every eighth bit in the ciphertext in various dimensions. It is intended to find regularities arising from the byte-oriented design of AES. The second setup is sensitive to abnormal fluctuations in the distribution of 1's in the ciphertexts (where up to two complete ciphertext blocks are used for one state of the gambling test). Setup 2 is especially interesting in connection with highly-patterned plaintexts. One checks if AES will be able to compensate for an extreme lack of 1's in the plaintext.

#### 4. FINDINGS

AES performed very well. The empirical results support the hypothesis that AES is not only an excellent cryptographic algorithm, but also a very interesting nonlinear random number generator for stochastic simulation.

We were unable to find any statistical weaknesses. Most of the  $p$ -values were completely uncritical, no systematic trends popped up in the tests. In the overlapping serial test (Setup 1), the number of rejections at the 5% and 1% level of significance are 7 and 4, respectively. Compared to the total number 180 of  $p$ -values computed, this is absolutely uncritical. The corresponding numbers for the gambling test are 5 and 1 for a total number of 112  $p$ -values. There was only one  $p$ -value below 0.005, namely 0.0010 for the gambling test, mode RNG,  $d = 64$  and sample size  $2^{28}$ . Recomputing this test with several different seeds showed that the suspect  $p$ -value was an outlier obtained by chance. We repeated all our tests with different keys (including all-one keys and random keys) and obtained almost identical results.

The results indicate that AES pseudorandom numbers are indistinguishable from real random numbers with respect to bit-vectors up to length 16 and with respect to the distribution of 0's and 1's in bit-vectors up to length 256.

From the results for DIFF we conclude that AES is able to compensate for an extreme lack of 1's in the plaintext, and hence provides excellent diffusion in this case. The results for KCOUNT provide some confidence that the key schedule used for generating the internal keys for the encryption

rounds in AES does a good job. Similar keys do not yield correlated output sequences.

Although no theoretical equidistribution analysis is available for AES, the empirical findings suggest the use of AES in the Counter mode as a nonlinear high-speed random number generator with a long period, for the cross-evaluation of simulation studies. In particular, we consider AES a valuable tool for checking stochastic simulations where one suspects possible interference between a linear random number generator and the model under investigation.

AES has the additional advantage of providing for explicit splitting capabilities needed in parallel stochastic simulation. Parallel streams can easily be generated using mutually disjoint ranges of plaintext blocks.

#### ACKNOWLEDGMENT

This article was written in part while S. Wegenkittl participated in the “Workshop on Random Number Generators and Highly Uniform Point Sets” at the Centre de Recherches Mathématiques of the Université de Montréal in June 2002. We would both like to express our gratitude to the organizer, Pierre L’Ecuyer, and to his co-workers for their hospitality.

#### REFERENCES

- CORON, J. S. AND NACCACHE, D. 1998. An accurate evaluation of Maurer’s universal test. In *Proceedings of Selected Areas in Cryptography 98*. Lecture Notes in Computer Science, vol. 1556. Springer-Verlag, New York, 57–71.
- DAEMEN, J. AND RIJMEN, V. 2002. *The Design of Rijndael*. Springer-Verlag, New York.
- KNUTH, D. 1998. *The Art of Computer Programming*, Vol. 2, 3rd ed. Addison-Wesley, Reading, Mass.
- L’ECUYER, P. AND HELLEKALEK, P. 1998. Random number generators: Selection criteria and testing. In *Random and Quasi-Random Point Sets*, P. Hellekalek and G. Larcher, Eds. Lecture Notes in Statistics, vol. 138. Springer-Verlag, New York, 223–266.
- L’ECUYER, P. AND SIMARD, R. 1999. Beware of linear congruential generators with multipliers of the form  $a = \pm 2^q \pm 2^r$ . *ACM Trans. Mod. Comput. Simul.* 25, 3, 367–374.
- L’ECUYER, P. AND SIMARD, R. 2002. *TestU01: A Software Library in ANSI C for Empirical Testing of Random Number Generators*. Département d’Informatique et de Recherche Opérationnelle, Université de Montréal. Software user’s guide.
- L’ECUYER, P., SIMARD, R., CHEN, E. J., AND KELTON, W. D. 2002a. An object-oriented random-number package with many long streams and substreams. *Oper. Res.* 50, 6, 1073–1075.
- L’ECUYER, P., SIMARD, R., AND WEGENKITTL, S. 2002b. Sparse serial tests of uniformity for random number generators. *SIAM J. Sci. Comput.* 24, 2, 652–668.
- LEE, H. AND WEGENKITTL, S. 1997. Inversive and linear congruential pseudorandom number generators in empirical tests. *ACM Trans. Mod. Comput. Simul.* 7, 2, 272–286.
- MARSAGLIA, G. 1996. DIEHARD. <http://stat.fsu.edu/~geo/diehard.html>.
- MATSUMOTO, M. AND KURITA, Y. 1994. Twisted GFSR generators II. *ACM Trans. Mod. Comput. Simul.* 4, 3, 254–266.
- MATSUMOTO, M. AND NISHIMURA, T. 1998. Mersenne Twister: A 623-dimensionally equidistributed uniform pseudorandom number generator. *ACM Trans. Mod. Comput. Simul.* 8, 1, 3–30.
- MAURER, U. 1992. A universal statistical test for random bit generators. *J. Crypt.* 5, 89–105.
- MENEZES, A. J., VAN OORSCHOT, P. C., AND VANSTONE, S. 1997. *Handbook of Applied Cryptography*. CRC Press, Boca Raton.
- NIST 2001a. *Federal Information Processing Standards Publication (FIPS PUB) 197*. NIST, National Institute of Standards and Technology. Available from <http://csrc.nist.gov/publications>.

- NIST 2001b. *A Statistical Test Suite for the Validation of Random Number Generators and Pseudo Random Number Generators for Cryptographic Applications*. NIST, National Institute of Standards and Technology. Available from <http://csrc.nist.gov/rng>.
- PINCUS, S. AND SINGER, B. H. 1998a. Randomness and degrees of irregularity. *Proc. Natl. Acad. Sci. USA* 93, 2083–2088.
- PINCUS, S. AND SINGER, B. H. 1998b. A recipe for randomness. *Proc. Natl. Acad. Sci. USA* 95, 10367–10372.
- SHANNON, C. E. 1949. Communication theory of secrecy systems. *Bell Syst. Tech. J.* 28, 657–715.
- SPENCER, M. E. AND TRAVARES, S. E. 1984. A layered broadcast cryptographic system. In *Proceedings of CRYPTO83*, D. Chaum, Ed. Plenum Press, New York, 157–170.
- WEGENKITTTL, S. 1999. Monkeys, gambling, and return times: Assessing pseudorandomness. In *Proceedings of the 1999 Winter Simulation Conference*, P. Farrington, H. Nembhard, D. Sturrock, and G. Evans, Eds. IEEE Press, Piscataway, N.J., 625–631.
- WEGENKITTTL, S. 2001a. Entropy estimators and serial tests for ergodic chains. *IEEE Trans. Inf. Theory* 47, 6 (Sept.), 2480–2489.
- WEGENKITTTL, S. 2001b. Gambling tests for pseudorandom number generators. *Math. Comput. Simul.* 55, 1–3, 281–288.
- WEGENKITTTL, S. 2002. A generalized  $\phi$ -divergence for asymptotically multivariate normal models. *J. Multiva. Anal.* 83, 288–302.
- WEGENKITTTL, S. AND MATSUMOTO, M. 1999. Getting rid of correlations among pseudorandom numbers: Discarding versus tempering. *ACM Trans. Mod. Comput. Simul.* 9, 3, 282–294.

Received October 2002; revised February 2003; accepted March 2003